

The Optimal Interactive Space for Hand Controller Interaction in Virtual Reality

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ABSTRACT

The definition of the optimal interactive space for hand direct operation, is an universally beneficial but less explored issue in the virtual reality (VR) domain. Through a specifically designed target acquisition experiment, it was found that (1) a bent arm posture generated a higher operational speed and accuracy, as well as lower physical exertion than a stretched arm posture; (2) arm operation at the shoulder below position was less laborious but more efficient and accurate than that at the shoulder above position; (3) the individual characteristic of handedness influenced the controller interaction performance at left- and right- sided interactive spaces.

Keywords: hand controller interaction, Fitts' law, three-dimensional user interface, handedness, ergonomics evaluation.

Index Terms: CCS—Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

Hand motion sensing interaction, including the bare hand interaction and a controller-based interaction, is largely limited by the bio-mechanical structure of the user's upper limb and its kinematic features [1-3]. In a VR environment, interactive targets distribute in surrounding spaces and the user performs distinct arm postures to acquire targets at different positions. Earlier kinematic studies have revealed that arm movement by some postures are perceived to be more comfortable than by other postures [4, 5]. The previous research gained a general knowledge about the muscle load and user perceived fatigue in arm's movement and freehand operations, but concerned little or at least incompletely on the hand interaction speed and accuracy at different arm postures. More subtly classified arm postures at different distances and angles in horizontal and vertical directions, had seldom been investigated.

To fill the knowledge gap and more importantly to develop guidelines for designing highly efficient hand controller interactive 3DUIs in VR, we conducted an empirical experiment in which participants were recruited to complete a series of target pointing and selection tasks, which is a most fundamental but representative interaction task in a VR environment.

2 METHODS

2.1 Participants

40 volunteers (24 male and 16 female) aged 21 to 36 ($M = 28.5$, $SD = 4.15$) were recruited. All the participants had normal or

corrected-to-normal vision, without any body impairments; 7 of them were left-handed while other 33 were right-handed.

2.2 Apparatus

An Oculus Quest (2nd generation, Developer kit) and a Touch controller were respectively used as the HMD and the handheld controller. The Oculus Quest was connected to a workstation computer (Win10, 128 GB memory, and 4.0 GHz Intel 64-core processor). The computer was also connected to a 32-inch-monitor, which was provided for supervising the visual field in the HMD. One motion-sensing camera of ASUS Xtion PROTM was mounted on the top-center margin of the monitor, which was used to identify and track the user's skeleton joints. **Figure 1** shows the experimental setting and scenario.

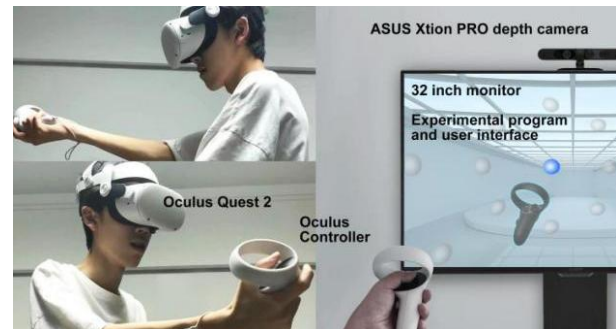


Figure 1: The experimental setting and scenario.

2.3 Procedure and Design

A specific target acquisition VR program was developed with the Unity3D engine in C#. In this program, a stereoscopic scene was rendered and the participant's shoulder joint of the operating arm was tracked and its position coordinates were calculated through the ASUS Xtion camera, a virtual controller showed in the VR scene to indicate the hand position in real time. The experimental task and the program interface was designed based on the Fitts' law pointing performance evaluation standard of ISO 9241-9 [6].

Table 1 presents the independent variables as follows.

Table 1: Independent variables in the experiment.

Independent Variables		
1. arm use choice	(1) left arm;	(2) right arm;
2. operating distance (the target towards the shoulder joint)	(1) large: 60 cm ~ 66 cm;	(2) middle: 46 cm ~ 51 cm;
	(3) small: 33 cm ~ 37 cm;	
3. angular movement amplitude	(1) 120°;	(2) 90°; (3) 60°; (4) 30°;
4. target size	(1) large: 4cm;	(2) middle: 2cm; (3) small: 1cm;
5. target direction	(1) rightward, R(→);	(2) right-upward, RU(↗);
	(3) upward, U(↑);	(4) left-upward, LU(↖);
	(5) leftward, L(←);	(6) left-downward, LD(↙);
	(7) downward, D(↓);	(8) right-downward, RD(↘);

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Each participant completed 18 task blocks. Each block consisted of 960 target acquisition trials. In each trial, the interval from the moment when the initial target was acquired to the moment when the target at the diagonal position was acquired was defined as the ‘movement time’ (MT); while acquiring the target, if the participant pressed the confirm button without keeping the controller hovering on the target, one acquisition error was counted and the target position was recorded by the program.

3 RESULTS

3.1 Target Acquisition Speed (MT Statistics)

Given the angular movement amplitude (α) and the angular target size (ω), the index of difficulty (ID) was calculated based on the updated formula of the Fitts’ law in [7]. It ranged from 2.51 to 7.07. A linear fitting of the MT statistics proved that the updated formula of Fitts’ law model had an acceptable applicability ($r^2 = 0.77$), but such result still had a gap from a perfect fitting. We then divided the MT statistics according to the target position and the arm choice, and then performed linear fitting respectively. **Table 2** presents the MT fitting result of the left arm and the right arm interaction at different directions, all had a high fitting degree ($\geq 90\%$).

Table 2: Fitts’ law linear fitting of two arms at different directions.

Arm Use Choice	Arm Movement Direction	Fitting Function	r^2
Left	R ($\rightarrow$)	$MT = -3560.8 + 1296 \times ID$	0.904
	RU ($\nearrow$)	$MT = -3674.5 + 1296 \times ID$	0.905
	U ($\uparrow$)	$MT = -3500.7 + 1272.9 \times ID$	0.900
	LU ($\nwarrow$)	$MT = -3143.5 + 1151.7 \times ID$	0.917
	L ($\leftarrow$)	$MT = -2380.1 + 861.9 \times ID$	0.901
	LD ($\swarrow$)	$MT = -2456.8 + 899.6 \times ID$	0.905
	D ($\downarrow$)	$MT = -2648.2 + 964.1 \times ID$	0.895
	RD ($\searrow$)	$MT = -2824.7 + 1024.1 \times ID$	0.909
Right	R ($\rightarrow$)	$MT = -2136.5 + 796.2 \times ID$	0.922
	RU ($\nearrow$)	$MT = -3010.1 + 1106.5 \times ID$	0.914
	U ($\uparrow$)	$MT = -3235.5 + 1194.6 \times ID$	0.919
	LU ($\nwarrow$)	$MT = -3465.1 + 1277.4 \times ID$	0.919
	L ($\leftarrow$)	$MT = -3319.4 + 1222.1 \times ID$	0.917
	LD ($\swarrow$)	$MT = -2536.3 + 945.3 \times ID$	0.917
	D ($\downarrow$)	$MT = -2413.6 + 896.0 \times ID$	0.919
	RD ($\searrow$)	$MT = -2321.3 + 856.9 \times ID$	0.919

A repeated-measures ANOVA proved that *Arm Choice* had a significant effect on the MT result ($F(1, 1199) = 481.68, p < 0.001$), right arm generated an overall smaller MT result than the left arm. *Operating Distance* also had a significant effect on the MT result ($F(2, 2398) = 1549.98, p < 0.001$), a bent arm posture generated a higher target acquisition speed than a stretched arm posture. In addition, *Target Direction* was also proven to have a significant effect on the MT result ($F(7, 8393) = 1063.70, p < 0.001$), arm movement and based target acquisition at the downward direction, i.e., D ($\downarrow$), had the smallest MT result; while that at the upward directions resulted in far larger MT results. There was a significant interaction effect of *Arm Choice* $\times$ *Target Direction* ($F(7, 8393) = 1373.36, p < 0.001$), indicated that human two arms respectively had a higher operational speed at the convenient side than that at the inverse side.

3.2 Target Selection Error Rate (ER)

A repeated-measures ANOVA showed that *Arm Choice* had a main effect on the ER result ($F(1, 39) = 59.90, p < 0.001$), the left arm operation generated a significantly lower error rate than that of the right arm operation ($M_{\text{left arm}} = 0.090, M_{\text{right arm}} = 0.094$); *Operating Distance* also had a significant effect on the ER result

($F(2, 78) = 647.77, p < 0.001$), the larger distance resulted in the higher ER result. Given the ER statistics at different directions, we then analyzed the *Target Direction* effect on the ER result. We compared the ER statistics at different directions in 3 conditions of ID, i.e., 2.51, 4.14 and 6.25, respectively. **Figure 2** shows the comparative result by each arm, in the left arm operation ER at the L ($\leftarrow$) is the lowest; and the ER at the left-sided directions, LU ($\nwarrow$), L ($\leftarrow$) and LD ($\swarrow$), were obviously lower than that at the right-sided directions, RU ($\nearrow$), R ($\rightarrow$) and RD ($\searrow$). In the right arm operation, the result was absolutely adverse.

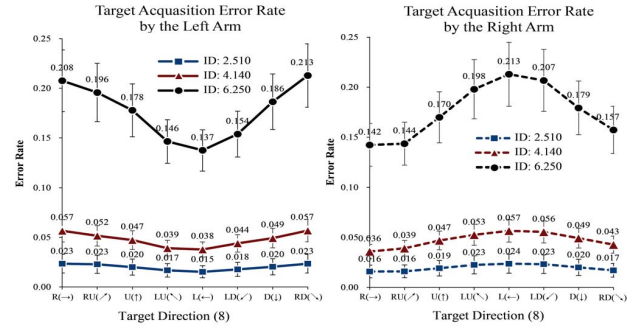


Figure 2: Target acquisition ER at different directions by two arms respectively.

4 CONCLUSION

There were mainly two findings and implications gained in the study: first, the arm posture in operation was closely correlated to the task completion speed and the controller operational accuracy, which implied guidelines for designing interactive space and distance in VR. Second, the arm operation having a larger angular deviation from the shoulder joint resulted in a poorer interaction speed and accuracy; and the arm operation at the shoulder below height was found to be more efficient and accurate than that at the shoulder above height, which implied more specific guidelines for designing user-friendly hand interaction user interfaces in VR.

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